

Source–Observation Riesz Channels

Spectral Projectors, Transfer Residues, and Rank-One Physical Modes

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Abstract

RH-194 finds that every surviving temporal root lies near one genuine base eigenvalue, but RH-192 forbids interpreting that root as the complete Frobenius eigenspace. This paper gives the exact source-relative object. For an isolated base spectral projector P , define

$$X_P = PS, \quad Y_P = P^*O^*.$$

These are the source and observation states cut out by the same Riesz channel. Their Frobenius pairing is

$$\langle Y_P, X_P \rangle_F = \text{tr}(OPS),$$

the contour residue of the scalar transfer function $g(z) = \text{tr}(O(zI - A)^{-1}S)$.

For pairwise disjoint spectral projectors $P_i P_j = \delta_{ij} P_i$, the cross pairing is diagonal:

$$\langle P_i^* O^*, P_j S \rangle_F = \delta_{ij} \text{tr}(OP_i S).$$

When the selected eigenvalues are simple and the residues are nonzero, residue normalization gives exact biorthogonal right/left eigenstates of the matrix left-multiplication dynamics. A 160-case complex nonnormal audit verifies projector resolution, cross-channel diagonality, transfer partial fractions, and normalized biorthogonality with zero failures.

The theorem explains the correct rank-one content of RH-194: one source–observation direction inside each repeated ambient eigenspace. A uniform residue lower bound, interval physical projectors, and all-level channel transport remain open.

1 Why a channel needs both source and observation

The source-cyclic space removes directions that cannot be reached from S . It may still contain modes invisible to the physical observation. A spectral packet intended for determinant or trace data must therefore be typed by both ends of the experiment.

Let $A \in \mathbb{C}^{n \times n}$, $S \in \mathbb{C}^{n \times m}$, and $O \in \mathbb{C}^{m \times n}$. The scalar transfer function is

$$g(z) = \text{tr}(O(zI - A)^{-1}S). \tag{1}$$

For large z it has the moment expansion

$$g(z) = \sum_{q \geq 0} \frac{\text{tr}(OA^q S)}{z^{q+1}}. \quad (2)$$

This is precisely the Frobenius source-observation transfer preserved by the RH-193 cyclic restriction.

2 Riesz projectors and matrix states

Let Γ be a positively oriented contour in the resolvent set of A and define

$$P = \frac{1}{2\pi i} \int_{\Gamma} (zI - A)^{-1} dz. \quad (3)$$

Then $P^2 = P$ and $PA = AP$. On the Frobenius space let $\mathcal{L}_A X = AX$.

Proposition 2.1 (Projected source channel). *The state*

$$X_P = PS \quad (4)$$

belongs to the source-cyclic space and satisfies

$$\mathcal{L}_A^q X_P = PA^q S \quad (q \geq 0). \quad (5)$$

The cyclic space generated by X_P is the part of the source orbit whose base spectrum lies inside Γ .

Proof. Since P is a polynomial in A in finite dimension, PS belongs to the span of powers of A applied to S . Commutation gives the orbit identity. $\square$

The dual state is

$$Y_P = P^* O^*. \quad (6)$$

It is generated by the adjoint dynamics and the observation seed.

3 The residue pairing

Theorem 3.1 (Channel pairing equals transfer residue). *For every Riesz projector P ,*

$$\boxed{\langle Y_P, X_P \rangle_F = \text{tr}(OPS)}. \quad (7)$$

Moreover

$$\text{tr}(OPS) = \frac{1}{2\pi i} \int_{\Gamma} g(z) dz. \quad (8)$$

Proof. Using $P^2 = P$,

$$\langle P^* O^*, PS \rangle_F = \text{tr}((P^* O^*)^* PS) = \text{tr}(OP^2 S) = \text{tr}(OPS).$$

Insert the Riesz integral and use linearity and cyclicity of the trace to obtain (8). $\square$

This identity is invariant under eigenvector scaling and does not require a choice of basis inside the spectral subspace.

4 Disjoint channels

Let $P_1, \dots, P_r$ be Riesz projectors for pairwise disjoint spectral sets. Then $P_i P_j = 0$ for $i \neq j$.

Theorem 4.1 (Cross-channel diagonality). *Put $X_j = P_j S$, $Y_i = P_i^* O^*$, and $r_i = \text{tr}(O P_i S)$. Then*

$$\langle Y_i, X_j \rangle_F = \delta_{ij} r_i. \quad (9)$$

Proof. Directly,

$$\langle Y_i, X_j \rangle_F = \text{tr}(O P_i P_j S) = \delta_{ij} \text{tr}(O P_i S).$$

□

Thus the observation automatically supplies the correct dual annihilation between distinct spectral channels. Poor conditioning can remain because $|r_i|$ may be small relative to $\|X_i\| \|Y_i\|$, but there is no off-diagonal algebraic contamination.

5 Simple eigenvalues

Suppose λ is simple, with right and left eigenvectors normalized by $w_\lambda^* v_\lambda = 1$. Then

$$P_\lambda = v_\lambda w_\lambda^*, \quad (10)$$

and

$$X_\lambda = v_\lambda (w_\lambda^* S), \quad (11)$$

$$Y_\lambda = w_\lambda (O v_\lambda)^*. \quad (12)$$

Both are single matrix vectors even though the full Frobenius eigenspace has dimension m .

Corollary 5.1 (Exact left/right eigenstates). *For a simple eigenvalue,*

$$\mathcal{L}_A X_\lambda = \lambda X_\lambda, \quad \mathcal{L}_A^* Y_\lambda = \bar{\lambda} Y_\lambda. \quad (13)$$

The scalar residue is

$$r_\lambda = (w_\lambda^* S)(O v_\lambda). \quad (14)$$

The mode is source-observable precisely when $r_\lambda \neq 0$. For a simple mode this implies both $P_\lambda S \neq 0$ and $P_\lambda^* O^* \neq 0$.

6 Residue-normalized biorthogonality

Choose simple disjoint eigenvalues $\lambda_1, \dots, \lambda_r$ with $r_i \neq 0$. Define columns in the Frobenius state by

$$V_i = X_i, \quad W_i = \frac{Y_i}{r_i}. \quad (15)$$

Theorem 6.1 (Exact source-channel packet). *The frames $V = [V_1, \dots, V_r]$ and $W = [W_1, \dots, W_r]$ satisfy*

$$W^*V = I_r, \quad (16)$$

and

$$\mathcal{L}_A V = V\Lambda, \quad \mathcal{L}_A^* W = W\Lambda^*, \quad W^* \mathcal{L}_A V = \Lambda, \quad (17)$$

where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_r)$.

Proof. Equation (9) gives $W^*V = I$. The eigenstate actions hold columnwise, and compression follows. $\square$

The packet has zero right and left residual. Its rank is the number of selected source-observable channels, not the full ambient Riesz rank.

7 Transfer partial fractions

If A has a complete simple spectrum, summing its rank-one projectors gives I and

$$(zI - A)^{-1} = \sum_j \frac{P_j}{z - \lambda_j}. \quad (18)$$

Taking the source-observation trace yields

$$g(z) = \sum_j \frac{r_j}{z - \lambda_j}. \quad (19)$$

Zero residues are exact pole cancellations in the scalar transfer function. The minimal source-observable realization keeps precisely the noncanceled Jordan data; in the simple case it keeps the indices with $r_j \neq 0$.

For a selected subset, equation (19) splits into the packet contribution and a complementary transfer remainder. This is a transfer decomposition, not a claim that the full Frobenius determinant factors with rank r .

8 Complex nonnormal identity audit

The implementation uses dimensions 2–9, source widths 1–4, and five independent trials per pair, giving 160 cases. Each test matrix is similar to a diagonal matrix with distinct complex eigenvalues and bounded similarity condition.

The audit checks:

1. resolution of the identity by computed projectors;
2. idempotence and cross-projector annihilation;
3. equality of channel pairing and residue;
4. diagonal cross-channel Gram matrix;
5. residue-normalized biorthogonality;
6. equality of the direct transfer solve and modal partial fractions.

All 160 cases pass the 10^{-9} aggregate tolerance.

9 Physical interpretation of RH-194

RH-194 finds four simple source-observable modes on each side at $\sigma = 0.01$ [1]. The present theorem upgrades their interpretation:

$$\text{one temporal root} \rightsquigarrow (P_\lambda S, P_\lambda^* O^*, r_\lambda).$$

This triple is basis-free and tied to the physical input-output experiment. It is the correct finite rank-one channel.

The raw residue normalization can be poorly conditioned. RH-196 constructs balanced coordinates on the same exact right/left spectral subspaces and proves their optimal cross-angle conditioning.

10 Boundary of the theorem

The Riesz projectors are exact finite objects when the contours are known. The physical projectors in RH-194 are presently computed by floating eigenvectors, not interval contour integration. The theorem also supplies no lower bound on $|r_\lambda|$ or its normalized overlap under refinement.

Finally, selecting a spectral quartet by first seeing its eigenvalues is not yet an intrinsic dynamical construction. Gate A requires a rule that finds and transports the channel without importing the desired spectral answer. Gates B–E and all claims concerning zeta zeros or RH remain untouched.

References

- [1] Liang Wang. Physical edge-root matching, 2026. Prime Dynamics Theory Program, RH-194.